In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv("StudentsPerformance.csv")

df.head()
```

Out[2]:

| | school | sex | age | address | famsize | Pstatus | Medu | Fedu | Mjob | Fjob | ... | famrel |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | GP | F | 18 | U | GT3 | A | 4 | 4 | at_home | teacher | ... | 4 |
| **1** | GP | F | 17 | U | GT3 | T | 1 | 1 | at_home | other | ... | 5 |
| **2** | GP | F | 15 | U | LE3 | T | 1 | 1 | at_home | other | ... | 4 |
| **3** | GP | F | 15 | U | GT3 | T | 4 | 2 | health | services | ... | 3 |
| **4** | GP | F | 16 | U | GT3 | T | 3 | 3 | other | other | ... | 4 |

5 rows × 33 columns

In [3]:
```
print("Dataset shape:", df.shape)
print("\nColumn names:")
print(df.columns)

print("\nDataset information:")
df.info()

print("\nStatistical summary:")
df.describe()
```

```
Dataset shape: (649, 33)

Column names:
Index(['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu', 'Fedu',
       'Mjob', 'Fjob', 'reason', 'guardian', 'traveltime', 'studytime',
       'failures', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery',
       'higher', 'internet', 'romantic', 'famrel', 'freetime', 'goout', 'Dalc',
       'Walc', 'health', 'absences', 'G1', 'G2', 'G3'],
      dtype='str')

Dataset information:
<class 'pandas.DataFrame'>
RangeIndex: 649 entries, 0 to 648
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      649 non-null    str
 1   sex         649 non-null    str
 2   age         649 non-null    int64
 3   address     649 non-null    str
 4   famsize     649 non-null    str
 5   Pstatus     649 non-null    str
 6   Medu        649 non-null    int64
 7   Fedu        649 non-null    int64
 8   Mjob        649 non-null    str
 9   Fjob        649 non-null    str
 10  reason      649 non-null    str
 11  guardian    649 non-null    str
 12  traveltime  649 non-null    int64
 13  studytime   649 non-null    int64
 14  failures    649 non-null    int64
 15  schoolsup   649 non-null    str
 16  famsup      649 non-null    str
 17  paid        649 non-null    str
 18  activities  649 non-null    str
 19  nursery     649 non-null    str
 20  higher      649 non-null    str
 21  internet    649 non-null    str
 22  romantic    649 non-null    str
 23  famrel      649 non-null    int64
 24  freetime    649 non-null    int64
 25  goout       649 non-null    int64
 26  Dalc        649 non-null    int64
 27  Walc        649 non-null    int64
 28  health      649 non-null    int64
 29  absences    649 non-null    int64
 30  G1          649 non-null    int64
 31  G2          649 non-null    int64
 32  G3          649 non-null    int64
dtypes: int64(16), str(17)
memory usage: 167.4 KB

Statistical summary:
```

Out[3]:

| | age | Medu | Fedu | traveltime | studytime | failures | famrel |
|---|---|---|---|---|---|---|---|
| **count** | 649.000000 | 649.000000 | 649.000000 | 649.000000 | 649.000000 | 649.000000 | 649.000000 |
| **mean** | 16.744222 | 2.514638 | 2.306626 | 1.568567 | 1.930663 | 0.221880 | 3.930663 |
| **std** | 1.218138 | 1.134552 | 1.099931 | 0.748660 | 0.829510 | 0.593235 | 0.955717 |
| **min** | 15.000000 | 0.000000 | 0.000000 | 1.000000 | 1.000000 | 0.000000 | 1.000000 |
| **25%** | 16.000000 | 2.000000 | 1.000000 | 1.000000 | 1.000000 | 0.000000 | 4.000000 |
| **50%** | 17.000000 | 2.000000 | 2.000000 | 1.000000 | 2.000000 | 0.000000 | 4.000000 |
| **75%** | 18.000000 | 4.000000 | 3.000000 | 2.000000 | 2.000000 | 0.000000 | 5.000000 |
| **max** | 22.000000 | 4.000000 | 4.000000 | 4.000000 | 4.000000 | 3.000000 | 5.000000 |

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
school        0
sex           0
age           0
address       0
famsize       0
Pstatus       0
Medu          0
Fedu          0
Mjob          0
Fjob          0
reason        0
guardian      0
traveltime    0
studytime     0
failures      0
schoolsup     0
famsup        0
paid          0
activities    0
nursery       0
higher        0
internet      0
romantic      0
famrel        0
freetime      0
goout         0
Dalc          0
Walc          0
health        0
absences      0
G1            0
G2            0
G3            0
dtype: int64
```

In [6]:
```
df.duplicated().sum()
```

Out[6]:
```
np.int64(0)
```

In [7]:
```
df.describe()
```

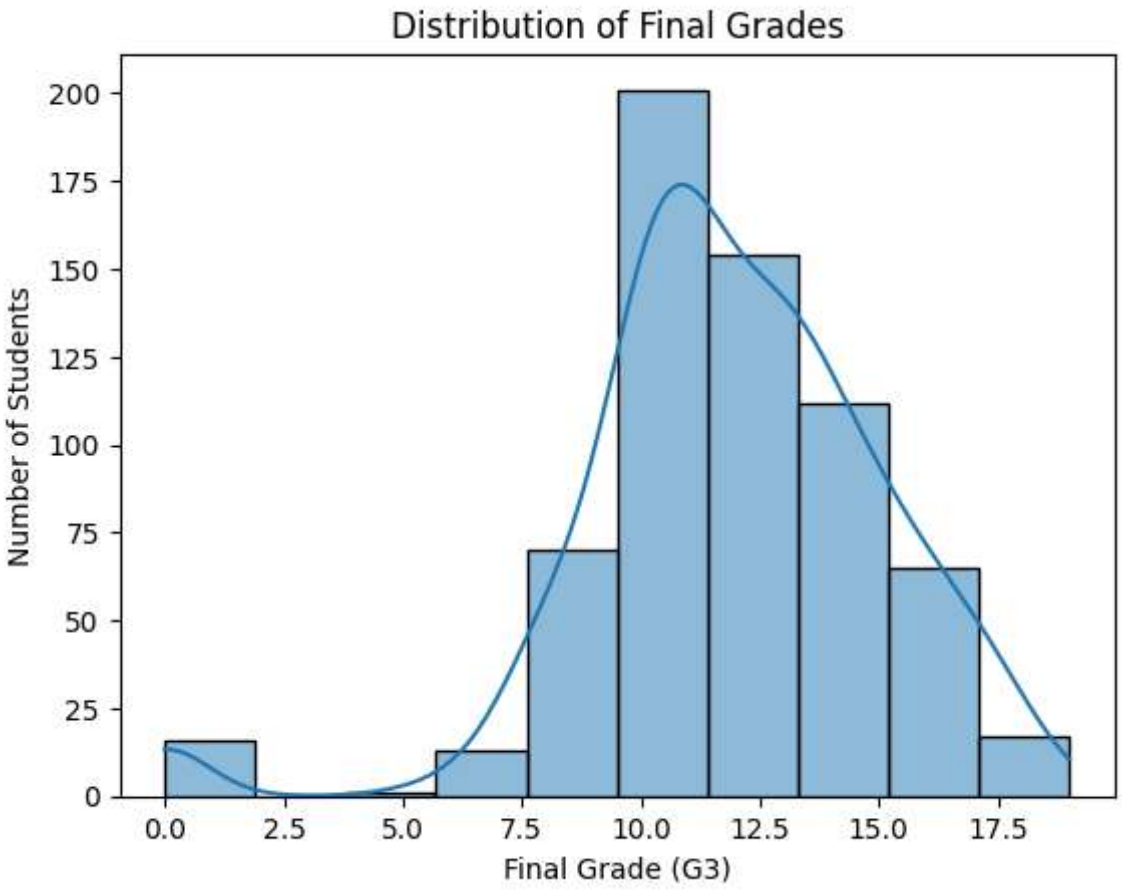
Out[7]:

| | age | Medu | Fedu | traveltime | studytime | failures | famrel |
|---|---|---|---|---|---|---|---|
| **count** | 649.000000 | 649.000000 | 649.000000 | 649.000000 | 649.000000 | 649.000000 | 649.000000 |
| **mean** | 16.744222 | 2.514638 | 2.306626 | 1.568567 | 1.930663 | 0.221880 | 3.930663 |
| **std** | 1.218138 | 1.134552 | 1.099931 | 0.748660 | 0.829510 | 0.593235 | 0.955717 |
| **min** | 15.000000 | 0.000000 | 0.000000 | 1.000000 | 1.000000 | 0.000000 | 1.000000 |
| **25%** | 16.000000 | 2.000000 | 1.000000 | 1.000000 | 1.000000 | 0.000000 | 4.000000 |
| **50%** | 17.000000 | 2.000000 | 2.000000 | 1.000000 | 2.000000 | 0.000000 | 4.000000 |
| **75%** | 18.000000 | 4.000000 | 3.000000 | 2.000000 | 2.000000 | 0.000000 | 5.000000 |
| **max** | 22.000000 | 4.000000 | 4.000000 | 4.000000 | 4.000000 | 3.000000 | 5.000000 |

In [8]:
```
df["G3"].mean()
```

Out[8]:
```
np.float64(11.906009244992296)
```

In [9]:
```
sns.histplot(df["G3"], bins=10, kde=True)
plt.title("Distribution of Final Grades")
plt.xlabel("Final Grade (G3)")
plt.ylabel("Number of Students")
plt.show()
```




In [10]:
```
df.describe()
```

Out[10]:

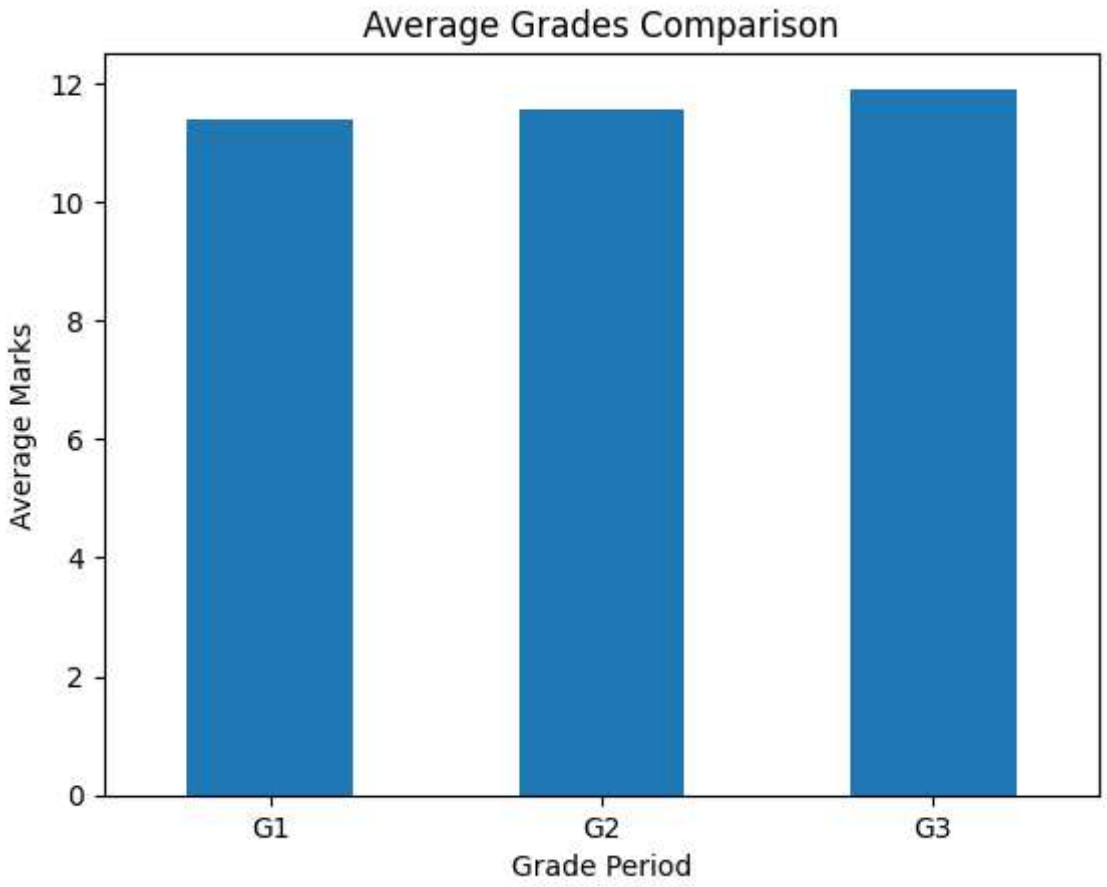
| | age | Medu | Fedu | traveltime | studytime | failures | famrel |
|---|---|---|---|---|---|---|---|
| **count** | 649.000000 | 649.000000 | 649.000000 | 649.000000 | 649.000000 | 649.000000 | 649.000000 |
| **mean** | 16.744222 | 2.514638 | 2.306626 | 1.568567 | 1.930663 | 0.221880 | 3.930663 |
| **std** | 1.218138 | 1.134552 | 1.099931 | 0.748660 | 0.829510 | 0.593235 | 0.955717 |
| **min** | 15.000000 | 0.000000 | 0.000000 | 1.000000 | 1.000000 | 0.000000 | 1.000000 |
| **25%** | 16.000000 | 2.000000 | 1.000000 | 1.000000 | 1.000000 | 0.000000 | 4.000000 |
| **50%** | 17.000000 | 2.000000 | 2.000000 | 1.000000 | 2.000000 | 0.000000 | 4.000000 |
| **75%** | 18.000000 | 4.000000 | 3.000000 | 2.000000 | 2.000000 | 0.000000 | 5.000000 |
| **max** | 22.000000 | 4.000000 | 4.000000 | 4.000000 | 4.000000 | 3.000000 | 5.000000 |

In [11]:
```
averages = df[["G1", "G2", "G3"]].mean()

averages.plot(kind="bar")

plt.title("Average Grades Comparison")
plt.xlabel("Grade Period")
plt.ylabel("Average Marks")
```

```
plt.xticks(rotation=0)
plt.show()
```



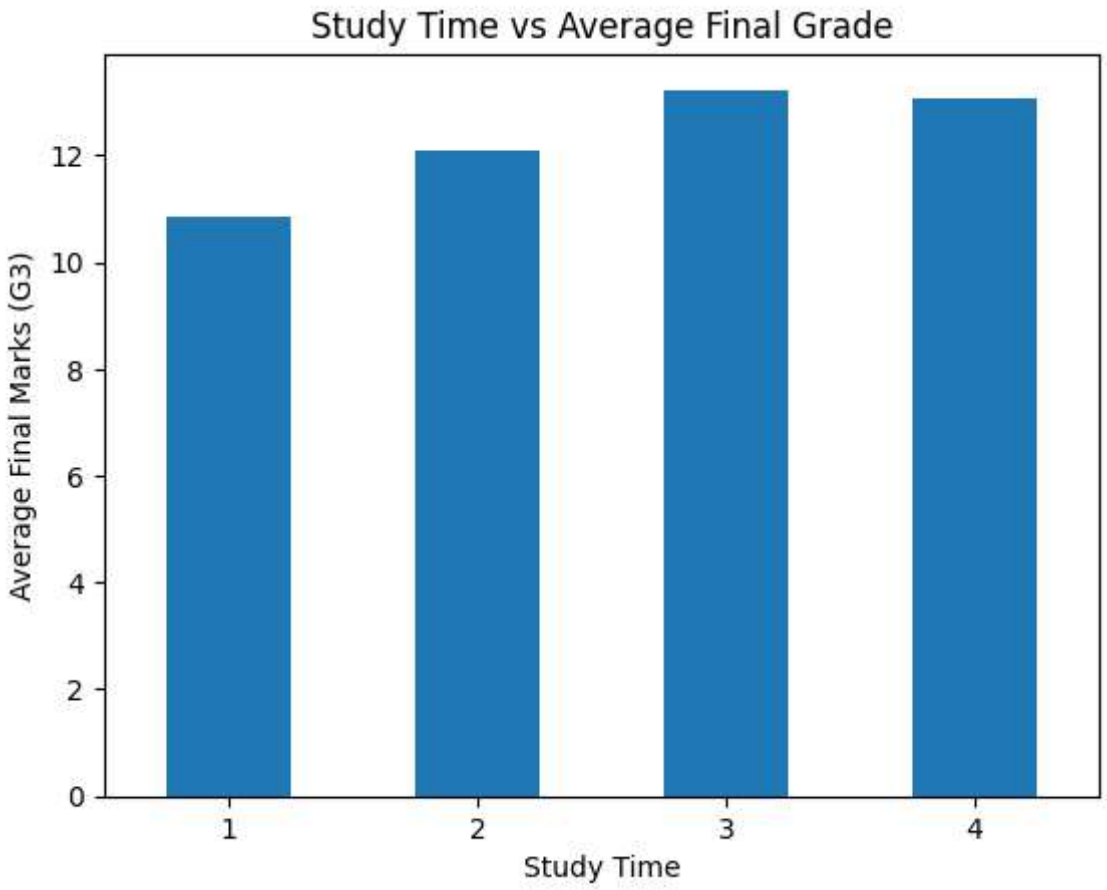

In [12]:
```
study_marks = df.groupby("studytime")["G3"].mean()

print(study_marks)

study_marks.plot(kind="bar")

plt.title("Study Time vs Average Final Grade")
plt.xlabel("Study Time")
plt.ylabel("Average Final Marks (G3)")
plt.xticks(rotation=0)
plt.show()
```

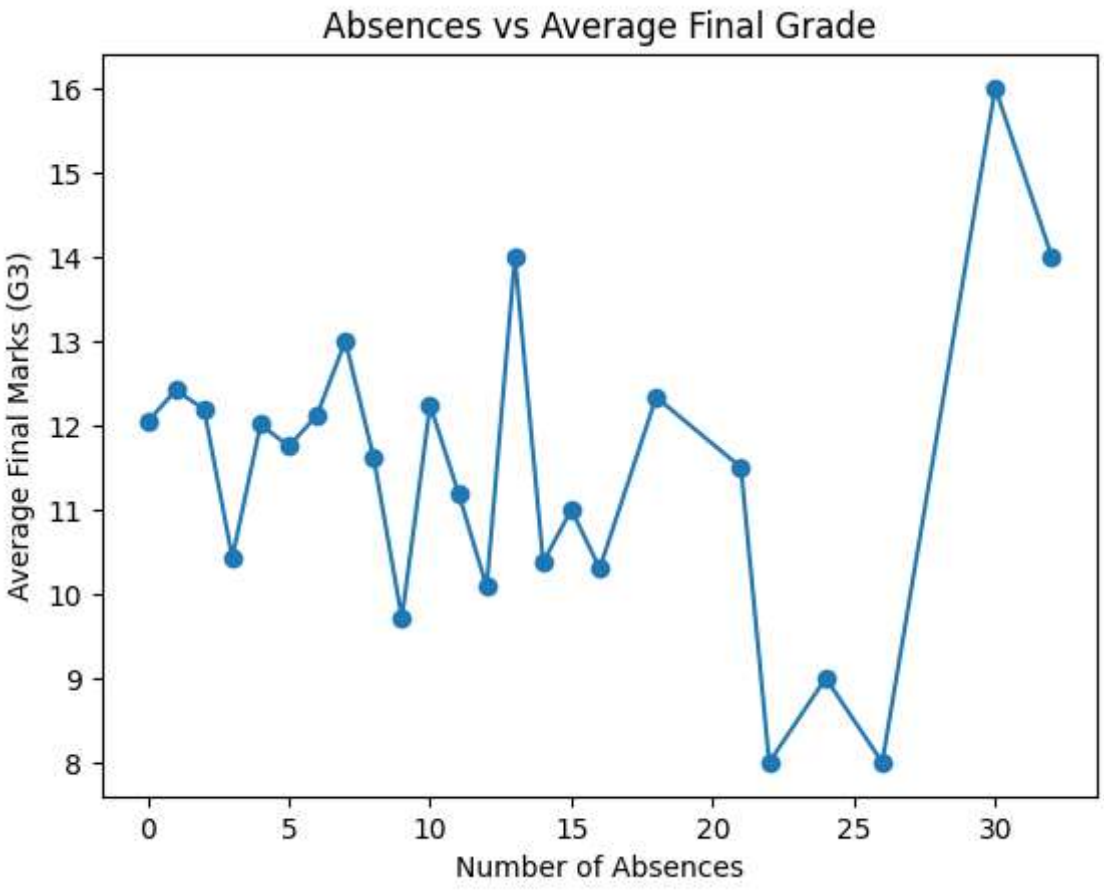
```
studytime
1    10.844340
2    12.091803
3    13.226804
4    13.057143
Name: G3, dtype: float64
```




In [13]:
```
absence_marks = df.groupby("absences")["G3"].mean()

absence_marks.plot(kind="line", marker="o")

plt.title("Absences vs Average Final Grade")
plt.xlabel("Number of Absences")
plt.ylabel("Average Final Marks (G3)")
plt.show()
```




In [14]:
```
correlation = df.corr(numeric_only=True)["G3"].sort_values(ascending=False)

print(correlation)
```

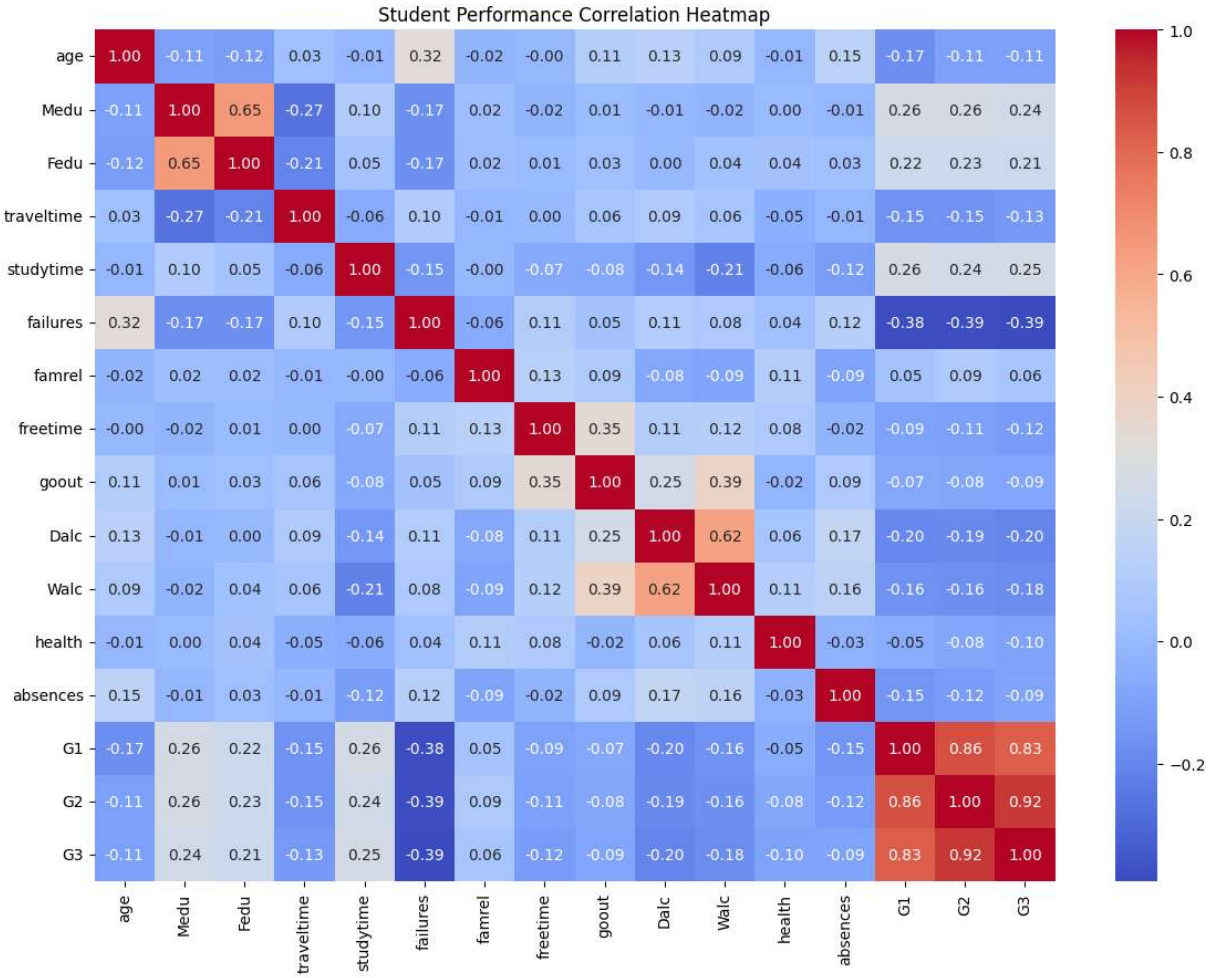
```
G3            1.000000
G2            0.918548
G1            0.826387
studytime     0.249789
Medu          0.240151
Fedu          0.211800
famrel        0.063361
goout        -0.087641
absences     -0.091379
health       -0.098851
age          -0.106505
freetime     -0.122705
traveltime   -0.127173
Walc         -0.176619
Dalc         -0.204719
failures     -0.393316
Name: G3, dtype: float64
```

In [15]:
```
plt.figure(figsize=(14, 10))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap="coolwarm",
```

```
    fmt=".2f"
)

plt.title("Student Performance Correlation Heatmap")
plt.show()
```



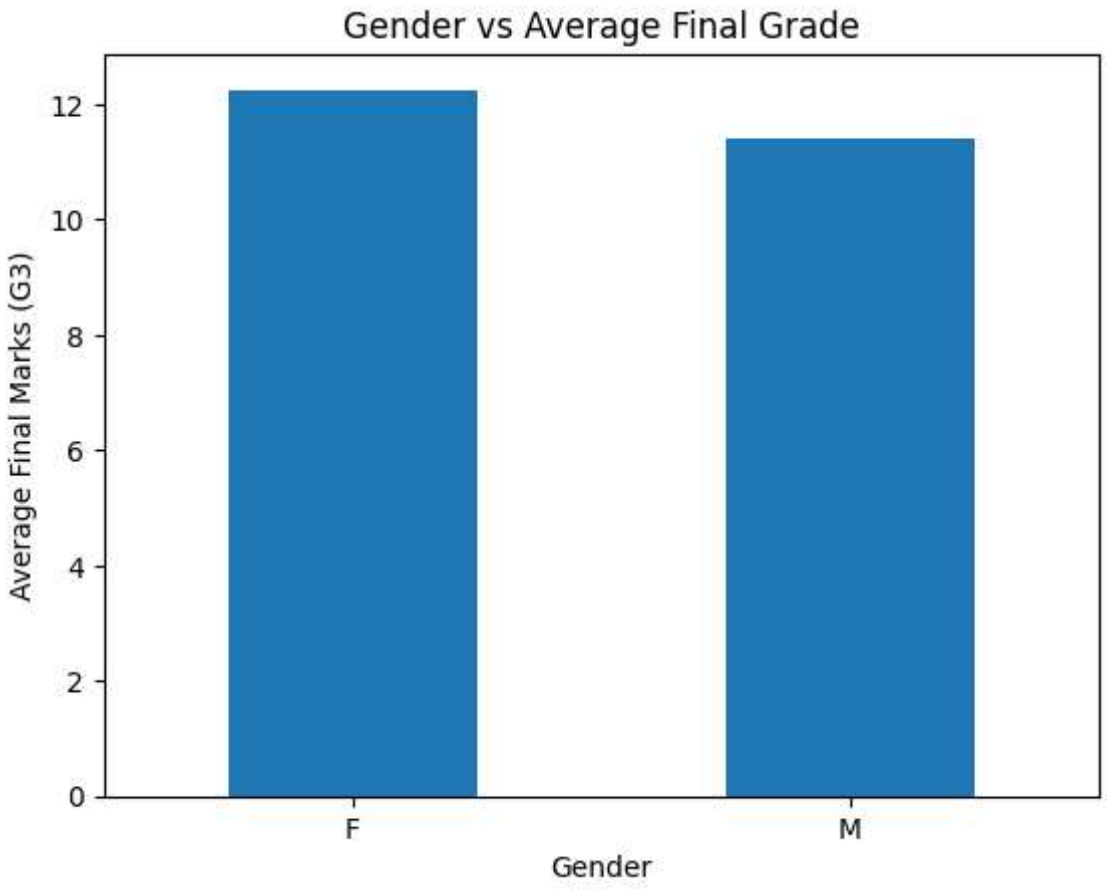

In [16]:
```
gender_marks = df.groupby("sex")["G3"].mean()

print(gender_marks)

gender_marks.plot(kind="bar")

plt.title("Gender vs Average Final Grade")
plt.xlabel("Gender")
plt.ylabel("Average Final Marks (G3)")
plt.xticks(rotation=0)
plt.show()
```

```
sex
F    12.253264
M    11.406015
Name: G3, dtype: float64
```



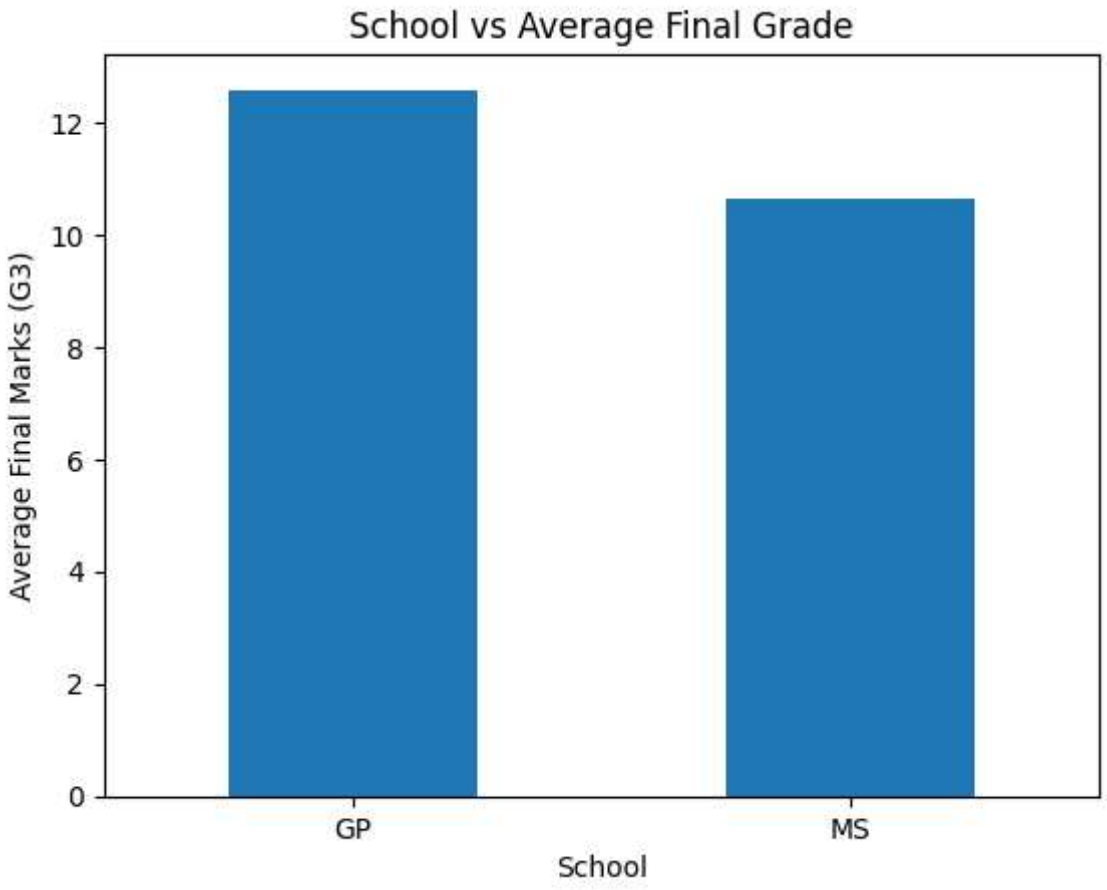

In [17]:
```
school_marks = df.groupby("school")["G3"].mean()

print(school_marks)

school_marks.plot(kind="bar")

plt.title("School vs Average Final Grade")
plt.xlabel("School")
plt.ylabel("Average Final Marks (G3)")
plt.xticks(rotation=0)
plt.show()
```

```
school
GP    12.576832
MS    10.650442
Name: G3, dtype: float64
```

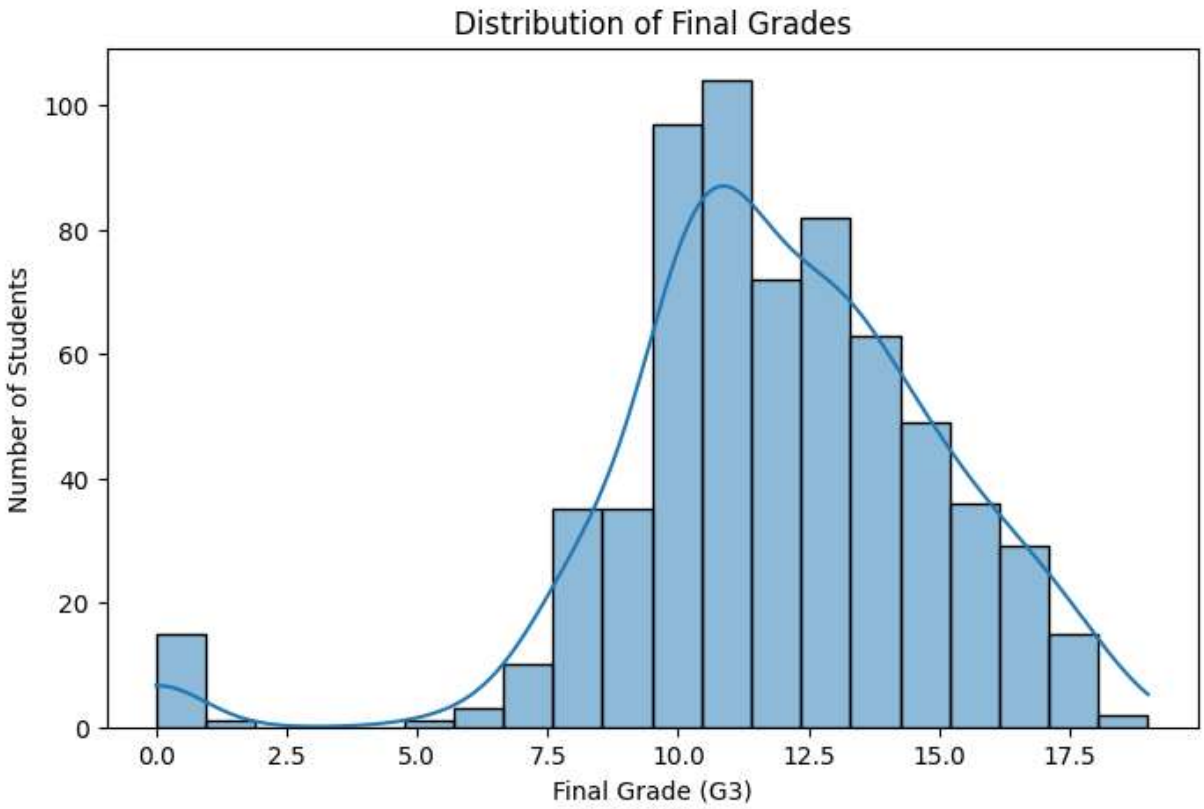



In [18]:
```
plt.figure(figsize=(8, 5))

sns.histplot(df["G3"], bins=20, kde=True)

plt.title("Distribution of Final Grades")
plt.xlabel("Final Grade (G3)")
plt.ylabel("Number of Students")

plt.show()
```


Distribution of Final Grades
Number of Students
Final Grade (G3)
0
20
40
60
80
100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

In [ ]: